

Universal Roughness and the Dynamics of Urban Expansion

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Urban sprawl reshapes cities, yet its quantitative laws remain elusive. Analyzing built-up expansion in 19 cities (1985–2015) with tools from surface growth physics in radial geometry, we reveal anisotropic, branchlike growth and a piecewise linear scaling between area and population. We uncover a robust local roughness exponent $\alpha_{\text{loc}} \approx 0.54$, coexisting with variable β and z . This unusual coexistence of universal and variable exponents offers a rare empirical test bed for nonequilibrium growth and an empirical basis for modeling urban sprawl.

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Urban sprawl, the expansion of cities into surrounding regions, has been extensively studied from demographic and economic perspectives, primarily focusing on population growth [1–3], migration patterns, and socioeconomic drivers [4–6]. Beyond population and economic drivers, urban expansion has led to environmental and sustainability challenges. The encroachment of urban areas onto agricultural land threatens food security, particularly in rapidly developing regions. Additionally, urban expansion near coastal zones increases the exposure of large populations to climate-related risks such as sea level rise and storm surges [5].

High-resolution satellite data reveal a sharp acceleration in global urban expansion: urban land cover grew by $\sim 80\%$ between 1985 and 2015, at a rate of $9687 \text{ km}^2/\text{year}$ [7], far exceeding earlier estimates [5]. Much of this growth has occurred in China, India, and Africa, driven by economic development and policy shifts, with GDP growth alone explaining nearly half the expansion in China [5].

Despite its fundamental importance, the spatial dynamics of built-environment expansion remain poorly quantified. Most analytical approaches to urban sprawl lack empirical grounding [8–12], with the notable exception of [13], which analyzed long-term patterns in London and Sydney. Many studies focus on descriptive land cover and population trends rather than quantitative models [5], and existing frameworks often prioritize economic or social factors over spatial dynamics [14,15].

A more systematic approach is needed to understand how cities grow [6], to identify whether universal principles

govern urban expansion, and—crucially—to determine the form of the evolution equation that describes urban sprawl, which remains an open question. Here, we introduce a new framework inspired by surface growth physics to study the spatiotemporal evolution of built-up areas. This framework, well established in systems such as turbulent liquid crystals [16,17], bacterial colonies [18], and tumor growth [19,20], and characterized by scaling laws and universality classes [21,22], has not yet been systematically applied to urban sprawl.

Key issues—Defining cities remains a key challenge [23]. Traditional definitions based on central flows [24] are subjective and limit quantitative analyses of spatial growth. A major advance came with the city clustering algorithm [25], which uses percolation theory to identify urban areas as clusters of built-up space. Wilson *et al.* [26] distinguished three urban growth types: infill (within existing fabric), expansion (at the fringe), and outlying growth, including isolated, linear, and clustered branch growth. These reflect interacting mechanisms of sprawl. We use the city clustering algorithm to extract the largest connected component—the giant cluster [27]—as a robust object to study urban growth.

A second key issue is the choice of an appropriate time parameter. Population P naturally serves as a clock for tracking urban evolution [4,6]. Unlike chronological time, it helps mitigate the effects of external disruptions—such as wars, epidemics, or short-term economic fluctuations—on growth dynamics. A quantitative analysis of urban sprawl should therefore examine the evolution of spatial variables (e.g., built-up area, average radius) as functions of P .

A third challenge is data availability. While population records span multiple decades, built-up area data is often

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incomplete or limited in coverage. Since our Letter requires both temporal built-up area and historical population data for urban areas, the data selection process is highly constrained. Additionally, we prioritized cities with relatively isotropic growth and minimal geographical constraints to enable a radial surface analysis. Despite these strong constraints, we successfully compiled a dataset covering most continents. We identified 19 major cities worldwide that meet these criteria, forming the basis of our analysis. Built-up area data was sourced from the WSF Evolution dataset [28], while population data was obtained from the World Population Prospects [29] [see Supplemental Material (SM) [30] for details].

Area growth speed and anisotropy—We begin by examining how total built-up area scales with population. Historical data from 1800 to 2000 [47] show that early urban areas largely coincided with the largest connected component (LCC). We identify piecewise linear trends with breakpoints indicating shifts in average population densities (see insets in Figs. 1 and S1 in SM [30]). In London, density dropped from 18 000 to 1800 inh/km² around 1914; in Paris, a similar decline occurred between 1979 and 1987, from 8500 to 810 inh/km². These findings contrast with cross-sectional scaling relations from [14] (see Fig. S2 in SM). From 1985 to 2015, we observe similar patterns across 19 cities (see Figs. S4 and S5 in SM), motivating the approximation $P \approx A P_{\text{tot}}/A_{\text{tot}}$, where P_{tot} is total population of the whole urban area A_{tot} , and P is the population of the LCC (and A its area). The relationship between A and population P reveals three growth patterns (Fig. 1): (a) linear scaling with constant density, as in Beijing (~ 3685 inh/km²); (b) piecewise linear with increasing density, as in Guatemala City; and (c) saturation, as in Las Vegas, where growth halts due to topographical or policy constraints [48,49].

Assuming isotropic growth with uniform density, $A \sim P$ implies $r(P) \sim P^{1/2}$, consistent with average trends [50]. To quantify anisotropy, we define $r(\theta, P)$ as the radius in a sector $[\theta, \theta + \delta\theta]$ centered on the city (see SM). We find

$r(\theta, P) \sim P^{\mu(\theta)}$, with $\mu(\theta) \in [0, 2.5]$ (see Fig. S6 in SM). Large exponents correspond to rapid growth along specific directions, often roads (see videos [46]). Deviations from the isotropic case $\mu = 1/2$ are used to quantify anisotropy (Fig. S7 in SM). Six cities exhibit low relative dispersion (e.g., Changzhou), while others such as Paris are highly anisotropic (see Figs. S8 and S9 in SM for maps of Changzhou and Kolkata).

Growth mechanisms—Our dataset reveals two distinct modes of surface growth: (i) *local growth*, occurring at the urban interface in a manner similar to diffusion, and (ii) *coalescence*, where expansion results from the merging of larger surrounding clusters (see growth videos [46]), as conjectured in [51]. To quantitatively distinguish these mechanisms, we write the new area that appears at time $t + 1$, denoted $\delta A(t + 1)$, as the sum of two terms,

$$\delta A(t + 1) = C_o(t) + C_n(t), \quad (1)$$

where C_o corresponds to clusters that already existed at time t (but were not connected to the largest connected component, LCC), and C_n denotes newly built areas that did not exist at time t .

To explore the relationship between growth patterns and urban expansion dynamics, we first plot (Fig. 2) the temporal averages (denoted by $\langle \cdot \rangle$) of both components in $\delta A(t + 1) = C_o(t) + C_n(t)$, normalized by $A(t)$, against the demographic pressure, quantified by the average population growth rate $g = \langle dP_{\text{tot}}/(P_{\text{tot}} dt) \rangle$. A high population growth rate is expected to exert pressure that accelerates urban expansion, which we indeed observe here. In addition, the inset of Fig. 2 shows the ratio $\langle C_o/C_n \rangle$ versus g , indicating that coalescence increases with demographic pressure and becomes predominant for $g > 10^{-2}$ (we show additional results in Fig. S10 in SM).

This trend is indeed confirmed in Fig. 2, where we observe that $\langle \delta A/A \rangle$ increases with g (a power-law fit suggests an exponent of approximately 1.22). When demographic pressure is low, surface expansion primarily

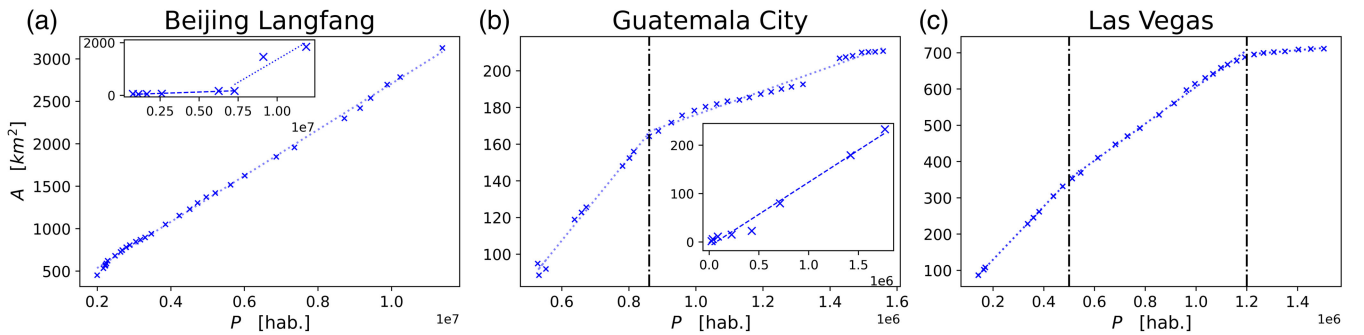


FIG. 1. Different growth patterns of the largest connected component (LCC) area vs population (1985–2015). (a) linear scaling at constant density (Beijing, ~ 3685 inh/km²); (b) piecewise linear growth with increasing density (Guatemala City, from 4437 to 15 314 inh/km²); (c) saturation plateau (Las Vegas, up to 16 782 inh/km²). Inset: historical trends (1800–2000). Beijing shows a density drop from 48 393 to 2882 inh/km² around 7×10^6 population, while Guatemala City maintains near-constant density (7576 inh/km²).

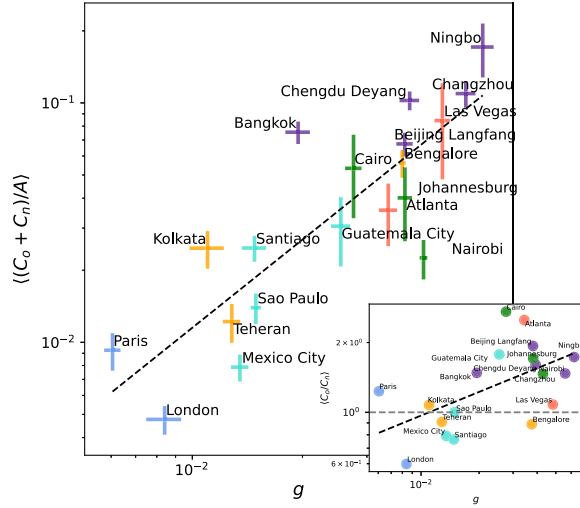


FIG. 2. Relative aggregated area as a function of demographic pressure, exhibiting a clear increasing trend. The dashed line represents a power-law fit with an exponent of approximately 1.22 (with $R^2 = 0.71$), provided as a visual guide. While the population growth rate changes from $\approx 1\%$ for London to 6% for Ningbo, the relative size of aggregates increases by more than 1.5 order of magnitude (the error bars represent the standard error). Inset: average ratio of coalesced built sites (C_o) relative to newly built sites (C_n). An increasing trend quantified by a Spearman's correlation of 0.51 is observed. The dark dashed line represents a power law of exponent 0.33. Color code: geographic region.

occurs through local agglomeration, with successive layer additions resembling surface growth processes in physics (as exemplified by Kolkata in Fig. S9 of SM). Conversely, under high demographic pressure, the LCC expands more abruptly by also assimilating large surrounding clusters—sometimes comparable in size to the LCC itself—exemplified by Ningbo, Las Vegas, and Cairo in Fig. S10 of SM. This mechanism closely parallels growth and coalescence patterns observed in animal dispersal [52,53].

Roughness exponents—In surface growth physics, it is standard practice to measure various exponents that characterize the evolution and roughness of interfaces, enabling the identification of distinct universality classes [21,22] (see Sec. V in [30] for a brief introduction). A well-established approach involves studying the width $w(L, t)$ of a one-dimensional growing surface over time for a system of linear size L , and a key result is that it follows the Family-Vicsek scaling relation $w(L, t) \sim t^\beta F(L/t^{1/z})$, where the scaling function behaves as $F(u \ll 1) \sim u^\alpha$. Here, β is the growth exponent, describing how quickly roughness develops, while α is the roughness exponent, characterizing the surface's spatial irregularity at macroscopic scale. The correlation length ξ evolves as $\xi \sim t^{1/z}$, where z is the dynamic exponent, indicating the rate at which roughness saturation is reached (when $\xi \sim L$, i.e., at crossover time $t_\times \sim L^z$). Most of these studies have focused on the one-dimensional case with a band geometry that is

not directly relevant here. Radial generalizations, more relevant to urban expansion, remain less explored but have been considered in [16,17,31,32,54–63]. In a radial geometry, where a surface expands on a two-dimensional plane from a central point, several issues arise. First, the system scale L used in traditional band geometry systems is not well-defined, requiring alternative methods [64], such as power spectrum collapse [31,33] or height-height correlation function [32], in order to determine the global roughness exponent α . Measuring growth via the total interface perimeter presents several challenges: first, obtaining a well-defined measure may require an ultraviolet cutoff for fractal boundaries [65]; second, its time evolution must be properly accounted for. If the size of the system grows at a faster rate than the correlations, saturation never occurs [56,64]. Second, anisotropy, often overlooked in previous studies, complicates growth characterization and makes it difficult to extend correlation-based or spectral methods [31,32,65].

In an isotropic system, the average radius, $\bar{r} = (1/2\pi) \sum_\theta r(\theta)$, effectively describes expansion. However, in the presence of anisotropy, this measure loses significance, complicating the task of measuring the growth exponent β or correlation functions. In order to remedy to this issue, a solution consists of introducing N sectors of size $\Delta\theta$, such that $N\Delta\theta = 2\pi$. The average width is then defined as an average over all sectors,

$$w^2(\bar{\ell}, P) = \frac{1}{N} \sum_{i=1}^N \langle [r(\theta, P) - \langle r \rangle_i]^2 \rangle_i, \quad (2)$$

where $\langle \cdot \rangle_i$ is the average computed over all angles in a given sector i ($\langle r \rangle_i$ is then the average radius over the sector i). For a given sector of average radius $\langle r \rangle_i$, we thus have an average arc length $\bar{\ell}(\theta) = \langle r \rangle_i \Delta\theta$ introducing in the system a new length scale.

In the isotropic case, $\langle r \rangle_i$ is independent of i , and the surface boundary is, on average, a circle. However, in the general anisotropic case, as seen in cities, the average interface deviates from a circular shape. We adopt the following anomalous scaling ansatz [33], as described in studies on anisotropic radial growth [64],

$$w(\bar{\ell}, P) = P^\beta F\left(\frac{\bar{\ell}}{P^{1/z}}\right), \quad (3)$$

where $\bar{\ell}$ is the average arc length of a sector of aperture $\Delta\theta$ defined above, and where $F(u \ll 1) \sim u^{\alpha_{\text{loc}}}$ with α_{loc} being the *local* roughness exponent (see SM for details). In our case, we do not expect the scaling to hold outside the $u \ll 1$ regime: as $\Delta\theta$ increases, the angular-resolved width will begin to reflect both curvature and anisotropy. The scaling form Eq. (3) suggests that plotting $wP^{-\beta}$ against $\bar{\ell}P^{-1/z}$ should result in a data collapse onto a single curve

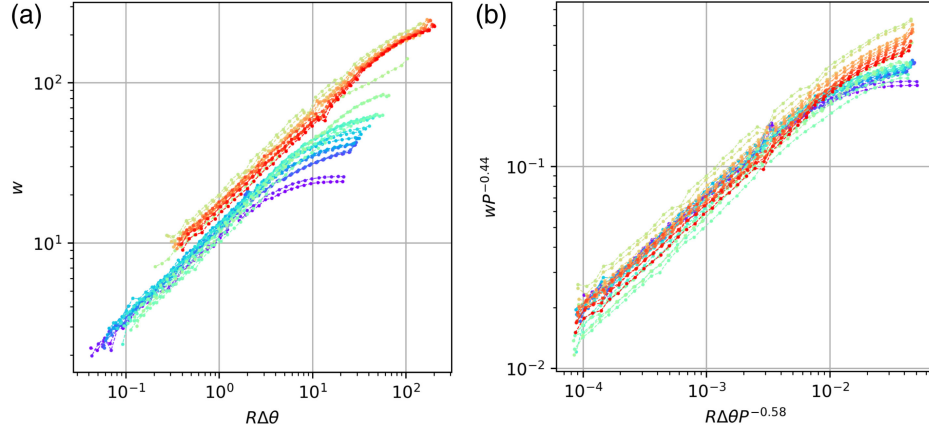


FIG. 3. (a) Interface width w , computed for angular sectors $\Delta\theta$, plotted against the average arc length $\bar{\ell} = R\Delta\theta$ (where $R = \langle r \rangle_i$ for each sector i) for the city of Ningbo, China, for each year between 1985 (purple) and 2015 (red). (b) According to the scaling form in Eq. (3), these curves should collapse onto a single master curve when plotting $wP^{-\beta}$ against $R\Delta\theta P^{-1/z}$. This collapse allows for the determination of the exponents β and z .

(for small enough angular aperture $\Delta\theta$), allowing for the determination of the exponents β and z (see SM for more details). We illustrate this method on the case of Ningbo (China) in Fig. 3. Once the collapse is achieved, in the first regime where $\bar{\ell}P^{-1/z} \ll 1$, we expect a slope equal to α_{loc} , following the scaling relation $w(\bar{\ell}) \sim \bar{\ell}^{\alpha_{\text{loc}}}$. A linear regression in log-log space allows us to determine the value of α_{loc} . For the cities analyzed in this Letter, we find a consistent local roughness exponent of $\alpha_{\text{loc}} = 0.54 \pm 0.03$, indicating minimal variation across cities. This value differs from the global roughness exponent α , estimated here via the relation $\alpha = \beta z$, which varies widely in the range $[0, 2]$. In nearly all cases—except for cities with $\beta \approx 0$ (and for Kolkata)—we observe $\alpha > \alpha_{\text{loc}}$, with an average value of $\alpha = 0.776 \pm 0.475$ (see Table S1 in [30] for details). This systematic difference indicates that the scaling is intrinsically anomalous: the standard Family-Vicsek scaling ansatz is not satisfied, and the interface does not fall into the super-rough regime [33].

The universality of this α_{loc} value suggests that the LCCs of cities share similar properties. Hence, the mechanisms governing local roughness appear fundamentally alike across all urban expansions studied here, despite the diversity of these cities and their growth dynamics. We note that the observed exponent close to $1/2$ could suggest a purely probabilistic origin, consistent with arguments in [34,35], although this point warrants further investigation to be fully elucidated (see Fig. S12 in SM for more details and discussions on this point).

The value of α_{loc} alone does not allow to identify any universality class and we have to measure β and z . In contrast, for these exponents our dataset reveals substantial variability in the exponents β and z across different cities (see Fig. 4 and Table S1 in SM). Figure 4 shows significant variability in the exponents β and z , allowing us to

distinguish broad city families according to their value of β and z . We also show the values corresponding to the usual universality classes considered in the statistical physics of surface growth [21,22] (see details in SM). Along the β axis, we can distinguish three broad families. The first group has low values, $\beta = 0.05 \pm 0.03$, indicating weak interface reactivity to population growth. These may signal a saturation regime, supported by small $1/z$ values suggesting halted correlation length growth. Alternative explanations include LCC densification (implying minimal

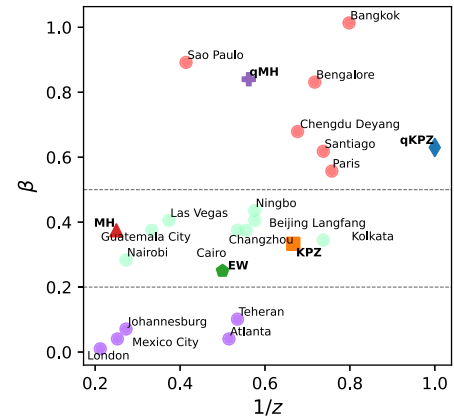


FIG. 4. Exponent β versus $1/z$ for cities in our dataset. We can distinguish three broad groups according to the value of β (shown in different colors): purple for β small, green for $\beta \approx 0.37$, and red for $\beta > 0.6$. We also indicate usual universality classes values for β and z : Edwards-Wilkinson (EW) with $\beta = 1/4$, $z = 2$ (and $\alpha = 1/2$), Kardar-Parisi-Zhang (KPZ) with $\beta = 1/3$, $z = 3/2$ (and $\alpha = 1/2$), and Mullins-Herring (MH) with $\beta = 3/8$, $z = 4$ (and $\alpha = 3/2$), along with two quenched universality classes at the depinning transition, qKPZ ($\beta = 0.63$, $z = 1$) and qMH ($\beta = 0.84$, $z = 1.78$).

area and roughness increase) or the absorption of similarly rough clusters. These cities generally experience low demographic pressure (except Atlanta). Within this group, z is large and highly variable (1.9–5). The second group, with $\beta \approx 0.37 \pm 0.04$, aligns with MH, EW, or KPZ universality classes, but z ranges broadly and uniformly from 1.4 to 3.7. The third group has $\beta > 0.5$ and more constrained z values ($1.25 < z < 1.43$), except for São Paulo ($z \approx 2.5$). These cities show faster roughness growth, likely due to high demographic pressure—though Paris and Santiago are exceptions. High β values may result from anisotropy and quenched noise, leading to local pinning and directional growth. Overall, no city fits neatly into the classical (EW, KPZ, MH) or quenched (qMH, qKPZ) universality classes. This could have been expected given the complex interplay of local growth, coalescence, and anisotropy of urban dynamics. Moreover, quenched noise affects importantly the behavior of growing interfaces [21,22,66] as (i) their characteristic exponents change with respect to their annealed counterpart, with typically rougher surfaces; (ii) breaks symmetries of the system (e.g., Galilean invariance $\alpha + z = 2$); and (iii) introduces pinning-depinning transitions, at which interfaces progress between pinned phases via avalanches, also modifying the exponents with respect to the moving interface phase (far from the depinning transition).

Conclusion—Through this quantitative analysis of urban sprawl, we identified key patterns. Built-up areas generally scale linearly with population, with density breakpoints marking transitions between densification and expansion regimes. Anisotropy can lead to branchlike growth, deviating from the average $r \sim \sqrt{P}$ behavior. The expansion of the largest connected component depends on demographic pressure: low pressure yields smooth, diffusionlike growth, while high pressure leads to coalescence-driven dynamics. Cities exhibit intrinsic anomalous roughening, quantified by a universal local roughness exponent $\alpha_{\text{loc}} \approx 0.5$, suggesting similar mechanisms across cities. The growth exponent β (and z) varies widely, indicating that none of the standard growth equations apply, but instead revealing three distinct city families classified by the value of β . These results show that both local growth and coalescence are required to accurately describe urban sprawl, with additional crucial impact from anisotropy and quenched noise.

Beyond scaling exponents, the full distribution of fluctuations is also of interest. The normalized radius does not follow a Tracy-Widom distribution, as observed in KPZ-type growth, but instead appears to follow a stretched exponential with an exponent close to 1/2 (see Sec. 13 of SM for details). Further investigation is needed to clarify the origin of this behavior.

This empirical characterization provides a necessary foundation for the mathematical modeling of urban sprawl: any theoretical model must reproduce these findings, which

serve as essential benchmarks and guides for understanding and simulating urban expansion.

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U. M. and R. G. collected the data. U. M., O. A., and R. G. processed the data. U. M. and O. A. developed the visualization code. U. M., R. G., and M. B. designed the study. U. M. and M. B. analyzed and interpreted the data and wrote the original draft. All authors interpreted and discussed the results, and reviewed and edited the manuscript.

The authors declare they have no conflicts of interest.

Data availability—The data that support the findings of this article are openly available [28,29,47].

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